



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT3007
Course title (English)	Optimization
Course title (Chinese)	最优化
Units	3
Language of Instruction	English
Description (English)	This course will focus on the basics of optimization theories and applications. The first part of the course will be mainly about linear programming, including how to formulate a linear program, the simplex method and the duality theory. Then we will study nonlinear optimization, focusing on the optimality conditions and some useful algorithms. Integer programming will also be briefly discussed in this course. To take this course, knowledge of calculus is required, and knowledge of linear algebra is recommended.
Description (Chinese)	本课程的主要目的是让学生熟悉基本的优化方法和他们在各种实际问题中的应用。主要的内容包括线性规划，单纯形算法，非线性优化，最陡下降方法，对偶理论，敏感度分析，整数优化。选修本课程，微积分相关知识是必需，同时建议具有一定的线性代数基础。

2. Prerequisites / Co-requisites

A. Prerequisites

(MAT1001 Calculus I or MAT1003 Analysis I or MAT1011 Honours Calculus or MAT1010 Calculus for Economic Analysis I) and (MAT1002 Calculus II or MAT1004 Analysis II or MAT1012 Honours Calculus II or MAT3010 Calculus for Economic Analysis II)

B. Co-requisites



3. Learning Outcomes

(a) Introduction (optional)

(b) Expected learning outcomes

Upon completion of the course, students will have acquired the ability to:

- appreciate the role of optimization in real-life decision making process
- transform a verbal description of a decision problem into a meaningful mathematical model
- solve a mathematical model by various tools (by hand, using different kinds of software)
- appreciate how and why an optimization algorithm works
- pick the best way to solve a given optimization problem
- intelligently interpret the solution one gets from solving an optimization problem
- appreciate the complexity of the optimal decision making process

(c) Implications for learning activities

Different types of learning activities will be used to help students to achieve the specified learning outcomes. These activities mainly include lectures and tutorials.

(d) Implications for assessment

Students will be assessed by their performance in homework, midterm examination, final examination and course project.

4. Course syllabus

- Linear programming
- Simplex method
- Duality theory and sensitivity analysis
- Nonlinear programming
- Nonlinear programming algorithms
- Analysis of algorithms, including convergence and optimality theory
- Integer programming
- Branch-and-bound algorithm
- Usage of computer packages such as MATLAB and/or Python

5. Assessment Scheme



Component/ method	% weight
Homework	20
Midterm	40
Final	40

6. Grade descriptor

General

Grade	Description
A	Demonstrates a thorough knowledge and understanding of the principles of optimization; successfully applies optimization methods at a sophisticated level in a wide variety of contexts; successfully transforms a verbal description of a given decision problem to a meaningful mathematical model; makes generalizations and justifies conclusions; understands and interprets the solution obtained from solving an optimization problem; communicates optimization methods in a clear, effective and concise manner, using correct techniques, notation and terminology; demonstrates the ability to integrate knowledge, understanding and skills from different areas of the course; uses technology proficiently.
A-	Demonstrates a broad knowledge and understanding of the principles of optimization systems; successfully applies optimization methods in a variety of contexts; successfully transforms a verbal description of a given decision problem to a meaningful mathematical model; makes generalizations and justifies conclusions; understands and interprets the solution obtained from solving an optimization problem; communicates optimization methods in a clear and effective manner, using correct techniques, notation and terminology; demonstrates some ability to integrate knowledge, understanding and skills from different areas of the course; uses technology proficiently
B	Demonstrates a good knowledge and understanding of the principles of optimization; successfully applies optimization methods in performing routine tasks; successfully transforms a verbal description of a given decision problem to a meaningful mathematical model in routine situations; understands the solution obtained from solving an optimization problem; communicates optimization methods using suitable notation and terminology; demonstrates an awareness of the links between different areas of the course; uses technology appropriately.
C	Demonstrates a satisfactory knowledge and understanding of the principles of optimization; successfully transforms a verbal description for some routine problems to a mathematical model; successfully carries out mathematical techniques to analyze and interpret the performance of a communication system in straightforward contexts; has limited ability to understand the solution obtained from solving an optimization problem; communicates optimization methods adequately; uses some appropriate notation and terminology; uses technology at a satisfactory level.
D	Demonstrates partial knowledge and limited understanding of the principles of



	optimization; attempts to transform a verbal description for some routine problems to a mathematical model; communicates some optimization methods; uses appropriate notation and terminology; uses technology to a limited extent.
F	Demonstrates minimal knowledge of the principles of optimization; demonstrates little or no ability to apply optimization methods in routine tasks; is unable to make effective use of technology.

7. Feedback for evaluation

Formal Course and Teaching Evaluation

Informal feedback to instructor and/or teaching assistant

Departmental retreat and programme review

8. Reading

9. Course components

Activity	Hours/week
Lectures	3
Tutorials	1

10. Indicative teaching plan

Week	Content/ topic/ activity
1	- Introduction to optimization
2	- Basics about linear optimization
7	- Interior point method, review, and midterm exam
3-5	- The simplex method
5-6	- Duality theories for linear programs and sensitivity analysis
8-10	- Introduction to nonlinear optimization - Optimality conditions for nonlinear programs - Convexity and convex programs
11-13	- Algorithms for nonlinear optimization problems
13-14	- Integer programs and their solution algorithms

11. Any other information



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12. Version date

Version number	10
As of (date)	2024/11/18